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MTRE3610: MODELING AND FEEDBACK CONTROL OF DYNAMIC SYSTEMS

FINAL PROJECT

DRONE DEPLOYMENT SYSTEM

by

GROUP 03

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EXECUTIVE SUMMARY

This report introduces a design for a control system, where an angled arm is attached to a backpack. The objective of the electric motor-powered arm will be to move up and down from the backpack and potentially reach a drone, and work as a retrieval zone for the drone. The drone will approach the backpack, and the robotic arm will grasp it. This report will focus on the raising and lowering of the arm, where a closed-loop control system is designed to ensure the arm will move at a reasonable speed to its desired destination while maintaining the weight applied.

A simple open loop method was performed at first. Although, the goal of the system was hindered by this design. If it was open-loop, the operator either had to be the wearer of the backpack or an observer to control the arm. Then they would have to move, adjust, and move again until the arm was at the desired position. This is not optimal.

The control system was modeled in SIMULINK to observe its performance. The SIMULINK model designed the full-nonlinear model to take the voltage input from the arm, go through integral actions, and a feedback loop accounting for the different forces in. The transfer function of the system had to exclude the mass of the arm so it could be created. The specifications of the system were implemented within SIMULINK and deviations from plant parameters were observed and noted. This demonstrated that the design was stable and durable, showing reliable movement and so ready for field implementation and evaluation as the next step in its progression.

TABLE OF CONTENTS

1. INTRODUCTION	
2. PROBLEM & OBJECTIVE STATEMENT	
2.1 ASSUMPTIONS AND SYSTEM CONSTRAINTS.....	
2.2 PERFORMANCE SPECIFICATIONS	
3. MODEL OF THE SYSTEM.....	
3.1 EQUATIONS OF MOTION.....	
3.2 TRANSFER FUNCTION	
3.3 OPEN-LOOP RESPONSE EVALUATION	
4. CONTROL SYSTEM DESIGN.....	
5. PERFORMANCE EVALUATION	
5.1 AS DESIGNED	
5.2 PERTURBATION & ROBUSTNESS EVALUATION	
6. CONCLUSIONS	
7. REFERENCES	
APPENDIX	
A.1 ADDITIONAL DATA	
A.2 ADDITIONAL DATA	

1. INTRODUCTION

A sketch of the mechanism is depicted in Figure 1.1. The arm is turned by an electric motor via a gearbox. Position 1 represents the arm in the “resting” position where it stays when there is either no drone or the drone has already landed. Position 2 represents the arm in the “ready” position where the drone will land on the platform and then return to position 1 for storage. The arm will begin at $\theta = 30$ degrees and will settle to $\theta = 120$ degrees. However, it will have a physical minimum and maximum angle constraint of 15 and 135 degrees to allow for safe overshoot and ensure safety during operation. The motor will be supplied with sufficient voltage in order for the arm to raise and lower at a reasonable rate. The arm will be designed to maintain and control the same angle position with and without the drone mass attached. The controller will also be designed to be robust enough to withstand the additional inertia that adding the drone mass introduces. Additionally, the drone is assumed it will land and rigidly attach itself onto an independently leveled platform which is controlled by an outside system.

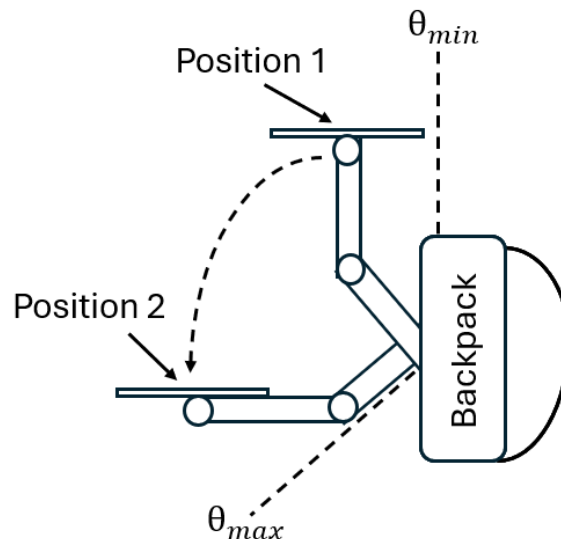


FIGURE 1.1 A SIDE VIEW OF THE SYSTEM SHOWING THE ARM AT A BENT ANGLE. THE ANGLE IS NEGLIGIBLE IN THE ANALYSIS.

2. PROBLEM & OBJECTIVE STATEMENT

The problem at hand is to control the movement of the electromotor's arm from θ_1 to θ_2 by adjusting the input voltage and analyzing the resulting angle output. The objective is to develop a control system that completes this task within a specified time t_s , while meeting the given design constraints and performance specifications.

a. ASSUMPTIONS AND SYSTEM CONSTRAINTS

For simplicity, this report will focus only on the operation of the arm movement. It is assumed that the friction is negligible (air drag or dry static/kinetic friction) within this operation and would be covered by the dampening coefficient. Additionally, since the mass of the drone could not be factored into the transfer function, it has been neglected and will be revisited in the closed system using Simulink. The moments of inertia, J , were calculated using the equation $J = \frac{1}{3}m \cdot r^2$ (for the rod) and $J = m \cdot r^2$ (for the drone mass), the dampening coefficients, b , were found by assuming values for the stall torque and no-load speed. Finally, it is also assumed that parameter changes are bounded and known as indicated in Table 2.1. Constraints on the system are arranged in Table 2.2.

TABLE 2.1 Drone Deployment system parameter values.

Parameter	Variable	Value	Tolerance	Units
Gear ratio	η	1:100	-	-
Resistor	R	10	$\pm 5\%$	ohms
Inductor	L	2	$\pm 5\%$	mH
Length of arm	l	50	-	cm
Mass of arm	m	0.5	$\pm 0.5\%$	kg
Radius of arm	r	0.5	$\pm 0.5\%$	cm
Moment of inertia of Shaft 1	J_1	1.2e-4	$\pm 10\%$	$Kg \cdot m^2$
Moment of inertia of Arm and Shaft 2	J_2	0.08723	$\pm 10\%$	$Kg \cdot m^2$
Dampening Coefficient of Motor	b_1	0.000019	$\pm 30\%$	N·m·s/rad
Dampening Coefficient of Arm	b_2	0.0001188	$\pm 30\%$	N·m·s/rad
Motor torque constant	K_m	0.062	$\pm 10\%$	N·m/A
Back EMF constant	K_b	0.062	$\pm 10\%$	V·s/rad

TABLE 2.2 System constraints.

Parameter	Variable	Value	Units
Maximum voltage of Motor	V_{max}	+25	<i>volts</i>
Minimum voltage of Motor	V_{min}	-25	<i>volts</i>
Maximum rotation of Arm	θ_{max}	135	<i>degrees</i>
Minimum rotation of Arm	θ_{min}	15	<i>degrees</i>
Vertical Start Position of Platform	$\theta_{position\ 1}$	30	<i>degrees</i>
Vertical Stop Position of Platform	$\theta_{position\ 2}$	120	<i>degrees</i>
Maximum weight of Drone	W_{max}	2	<i>kg</i>

b. PERFORMANCE SPECIFICATIONS

The controlled system must exhibit transient and steady-state behaviors that meet the performance specifications outlined in Table 2.3 each time the arm moves to a new position.

TABLE 2.3 Performance specifications.

Parameter	Variable	Value	Units
Maximum steady state position error	E_{ss}	0	degrees
Maximum position overshoot	M_p	15	%
Maximum settling time	t_s	1	s

3. MODEL OF THE SYSTEM

This system can be modelled as an electro-mechanical system as shown in Figure 2.1 below. In this case, the mass of the arm and drone are disregarded for the creation of the transfer function but will be reimplemented in the Simulink simulation. The input of the system will be a voltage, and the output will be the angle of the arm.

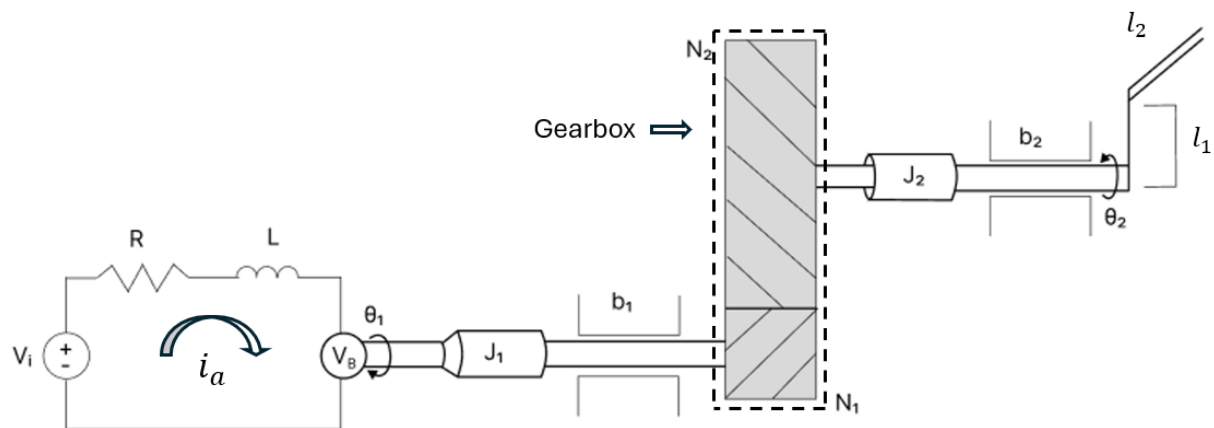


FIGURE 2.1 SKETCH OF THE ELECTRIC MOTOR SYSTEM, SHOWING THE TWO GEARS THAT DRIVE THE ARM.

3.1 EQUATIONS OF MOTION

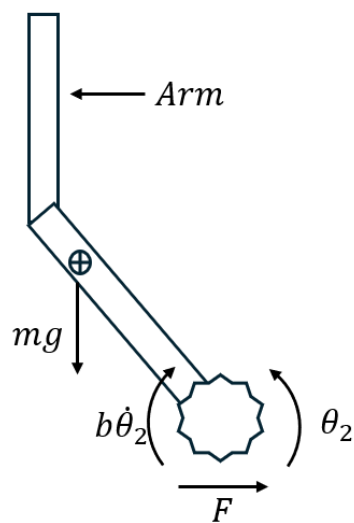


FIGURE 2.1 FREE BODY DIAGRAM OF THE ARM WITH GEAR.

The equations of motion are obtained by the free body diagram in Figure 3.1, considering the weight of the arm. θ_1 is taken from the motor, and θ_2 is taken from the arm, as pictured.

$$J_2\ddot{\theta}_2 = Fr_2 - b_2\dot{\theta}_2 - mgL \sin \theta_2 \quad (1)$$

$$J_1\ddot{\theta}_1 = T - b_1\dot{\theta}_1 - Fr_1 \quad (2)$$

$$V_i = R_a i_a + L \frac{di_a}{dt} + V_b \quad (3)$$

3.1 TRANSFER FUNCTION

To obtain the transfer function G_p , the three equations of motion will be combined. All transfer functions have initial values of zero, meaning when the Laplace is taken for the equations, any initial values from the Laplace formulas will equal to zero. Torque (T) from the motor has a relationship of being the equivalent of a constant from the DC motor circuit (K_m) multiplied by the current (i_a), and this is because of the windings that a DC motor has. Also, the back EMF voltage of the DC motor (V_b) is the equivalent of the back EMF constant multiplied by the angular velocity of the first shaft. The inductance, L , has a negligible value that makes it insignificant in the outcome of the system. It will be assumed to be equal to zero in Equation 2, and it will be reintroduced in the SIMULINK model.

$$T = K_m i_a \quad (4)$$

$$V_b = K_b \dot{\theta}_1 \quad (5)$$

$$J_2\ddot{\theta}_2 = Fr_2 - b_2\dot{\theta}_2 + mgL \sin \theta_2 \quad (6)$$

Substitute T and V_b .

$$J_1\ddot{\theta}_1 = K_m i_a - b_1\dot{\theta}_1 - Fr_1 \quad (7)$$

$$V_i = R_a i_a + K_b \dot{\theta}_1 \quad (8)$$

Create relationships between θ_1 and θ_2 .

$$r_1\theta_1 = r_2\theta_2 \quad (9)$$

$$r_1\dot{\theta}_1 = r_2\dot{\theta}_2 \quad (10)$$

$$r_1\ddot{\theta}_1 = r_2\ddot{\theta}_2 \quad (11)$$

$$\eta = \frac{r_2}{r_1} = \frac{N_2}{N_1} = \frac{\theta_1}{\theta_2} \quad (12)$$

$$\theta_1 = 2\theta_2 \quad (13)$$

Get F by itself in Equation 1 and plug it into Equation 7.

$$F = \frac{(K_m i_a - J_1 \ddot{\theta}_1 - b_1 \dot{\theta}_1)}{r_1} \quad (14)$$

Plug in Equation 14 into Equation 6.

$$J_2 \ddot{\theta}_2 = \frac{r_2}{r_1} (K_m i_a - J_1 \ddot{\theta}_1 - b_1 \dot{\theta}_1) - b_2 \dot{\theta}_2 + mgl \sin \theta_2 \quad (15)$$

Insert θ relationships, Equation 12 into Equation 15.

$$J_2 \ddot{\theta}_2 = \eta (K_m i_a - J_1 \ddot{\theta}_2 - b_1 \dot{\theta}_2) - b_2 \dot{\theta}_2 + mgl \sin \theta_2 \quad (16)$$

$$J_2 \ddot{\theta}_2 = \eta K_m i_a - J_1 \eta^2 \ddot{\theta}_2 - b_1 \eta^2 \dot{\theta}_2 - b_2 \dot{\theta}_2 + mgl \sin \theta_2 \quad (17)$$

The full non-linear model is given by the following, with V_i, i_a, θ_2 as variables.

$$V_i = R_a i_a + K_b \eta \dot{\theta}_2 \quad (18)$$

$$(J_2 + J_1 \eta^2) \ddot{\theta}_2 + (b_2 + \eta^2 b_1) \dot{\theta}_2 = \eta K_m i_a + mgl \sin \theta_2 \quad (19)$$

Cancel out the nonlinear component.

$$V_i = u - \frac{R_a}{\eta K_m} mgl \sin \theta_2 \quad (20)$$

$$i_a = \frac{u}{R_a} - \frac{mgl \sin \theta_2}{\eta K_m} - \frac{K_b \eta \dot{\theta}_2}{R_a} \quad (21)$$

$$(J_2 + J_1 \eta^2) \ddot{\theta}_2 + (b_2 + \eta^2 b_1) \dot{\theta}_2 = \eta K_m \left(\frac{u}{R_a} - \frac{mgl \sin \theta_2}{\eta K_m} - \frac{K_b \eta \dot{\theta}_2}{R_a} \right) + mgl \sin \theta_2 \quad (22)$$

$$(J_2 + J_1 \eta^2) \ddot{\theta}_2 + (b_2 + \eta^2 b_1) \dot{\theta}_2 = \frac{\eta K_m}{R_a} u - \frac{K_m K_b \eta^2}{R_a} \dot{\theta}_2 \quad (23)$$

$$(J_2 + J_1 \eta^2) \ddot{\theta}_2 + (b_2 + \eta^2 b_1) \dot{\theta}_2 + \frac{K_m K_b}{R_a} \dot{\theta}_2 = \frac{\eta K_m}{R_a} u \quad (24)$$

Take the Laplace of Equation 24. This is so the transfer function can be formed. Factor out θ_2 and U .

$$\theta_2 \left((J_2 + J_1 \eta^2) s^2 + (b_2 + b_1 \eta^2) s + \frac{K_m K_b \eta^2}{R_a} s \right) = \frac{\eta K_m}{R_a} U \quad (25)$$

Transfer Function:

$$G_p = \frac{\theta_2(s)}{U(s)} = \frac{\eta K_m}{(J_2 + J_1 \eta^2)(R_a)s^2 + (b_1 \eta^2 + b_2)(R_a)s + \eta^2 K_m K_b s} \quad (26)$$

In this equation, J_2 is the inertia of the arm and J_1 is the motor shaft's rotor inertia, b_2 and b_1 are the friction coefficients of the shafts, R_a is the resistance of the motor windings, L is the inductance of the motor, K_m is the motor torque constant, K_b is the back EMF constant, and η is the gear box ratio between θ_1 and θ_2 . The values corresponding to these parameters are found in Table 2.1. Since it is desired to measure θ_2 in degrees and V_i in volts, conversion factors are applied to get the correct units.

$$G_p = \frac{\theta_2(s)}{U(s)} = \frac{\eta K_m}{(J_2 + J_1 \eta^2)(R_a)s^2 + [(b_1 \eta^2 + b_2)(R_a) + \eta^2 K_m K_b]s} \quad (27)$$

$$= \frac{(100)(0.062)}{(8.723 \times 10^{-2} + 1.2 \times 10^{-4}(100^2))(10)s^2 + (1.9 \times 10^{-5}(100^2) + 1.188 \times 10^{-3})(10)s + (100^2)(0.062)(0.062)s}$$

$$G_p = \frac{\theta_2(s)}{V_I(s)} = \frac{6.2}{12.87s^2 + 40.35s} \quad (28)$$

Poles:

$$s_0 = 0, s_1 = -3.14 \quad (29)$$

Once the constants were plugged in and the equation was simplified, there was a 2nd order system left in the denominator. When the poles were solved for, they were all on the real axis.

3.2 OPEN-LOOP RESPONSE EVALUATION

The model of our system is implemented into MATLAB Simulink (Fig 3.1) to be able to simulate a voltage step input into our non-linear open loop system where the output is the angle position of the arm with the mass attached to the end of our arm. At $T = 5$ seconds, -25 volts were applied to the motor and then set back to 0 at $T = 9$ seconds. The response from the system (Fig 3.4) was as expected. For the first 5 seconds where 0 volts were applied, the arm fell from the minimum angle constraint of $\pi/12$ radians to the maximum angle constraint of $3\pi/4$ radians, due to gravity

acting on the center of gravity of the arm and no additional input. It falls slowly with no load due to the 1:100 gear ratio combined with the back EMF generated from rotating the motor with such a high gear ratio. At $T = 5$ seconds, the voltage of -25 is applied which makes the motor rotate the arm upwards till it hits the minimum angle constraint again because there is no controller regulating where it stops at. Then at $T = 9$ seconds the voltage is set to 0 again allowing the arm to fall back down again.

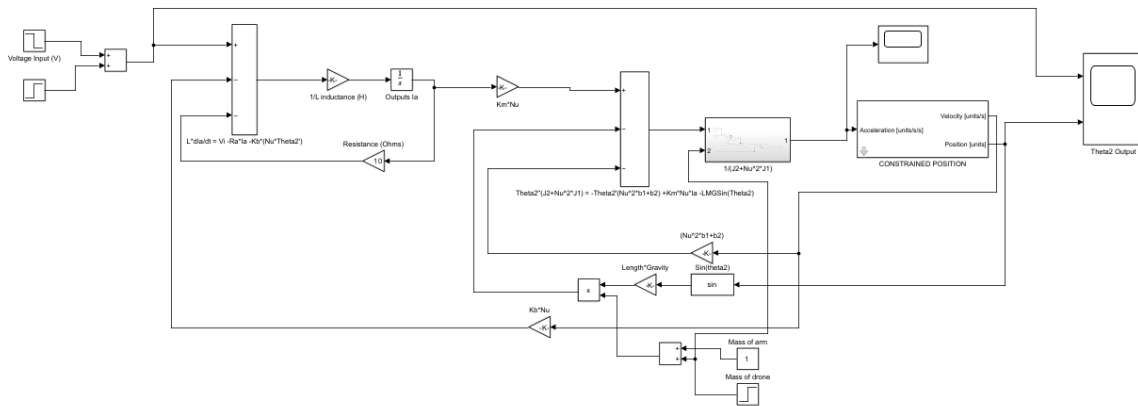


FIGURE 3.1 SIMULINK MODEL FOR OPEN LOOP SYSTEM

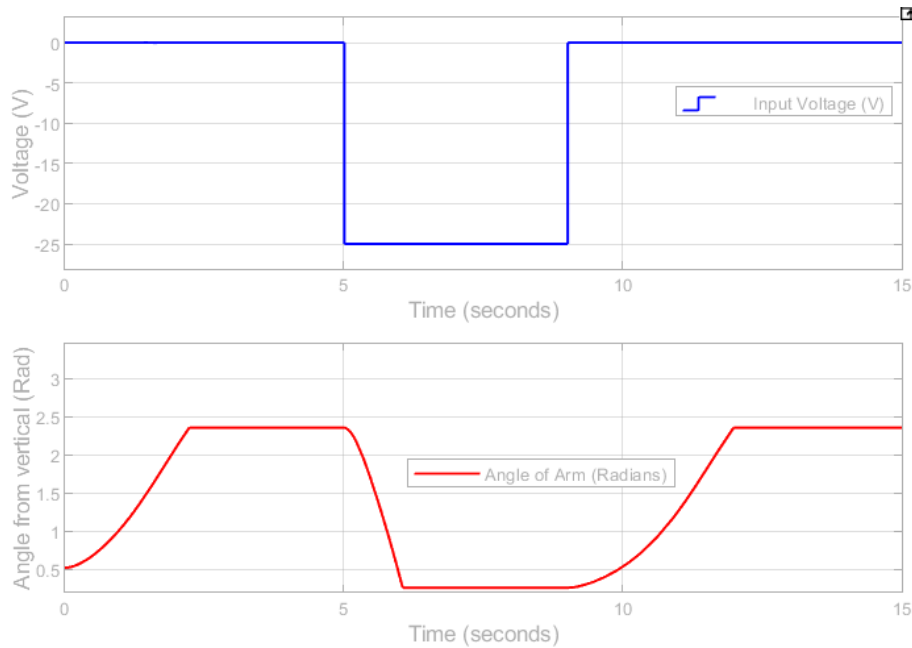


FIGURE 3.2 OPEN-LOOP STEP RESPONSE WITH NOMINAL PLANT PARAMETERS

4. CONTROL SYSTEM DESIGN

The closed-loop controller for the Drone-and-Arm system is designed via the root locus technique [1] with the aid of the MATLAB routine RLTOOL. The performance specifications call for a maximum steady state position error of 0%, maximum overshoot of 15% and a settling time of 1.0 s. These requirements give a range of values within the Real-Imaginary plane where the roots of the controller transfer function need to be to be able to meet them. This region is everywhere not shaded in yellow in figure 4.1 showing where we can place our gains. In our plant transfer function, there are no zeros in the numerator and two poles in the denominator located at -3.1352 as well as the other at the origin meaning there is a natural integrator in our system constituting the basis for our controller's design. The overall gains used for our controller design were chosen based on the locations of the closed-loop poles outside the shaded region. During performance evaluation, we tested our system against multiple variable conditions allowing us to find the best fit controller for our system.

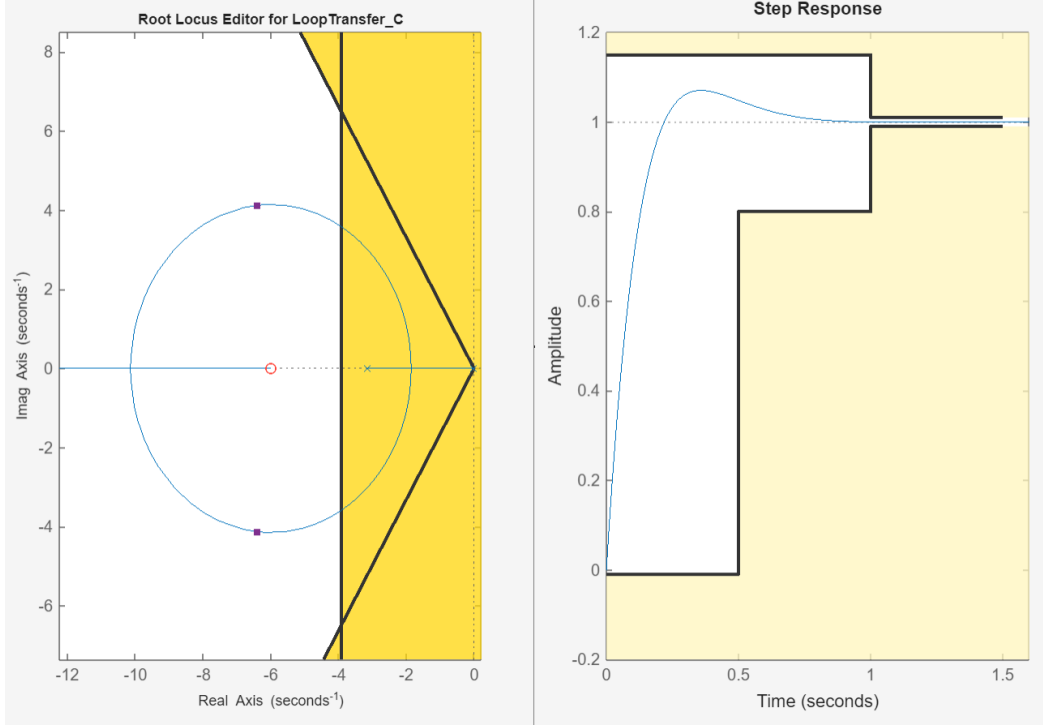


FIGURE 4.1 ROOT LOCUS AND RESPONSE CURVE FOR THE CONTROL SYSTEM WITH NO MASS.

The controller gains are extracted from

$$G_c = 20(s + 6) = K_D(s + K_P) = K_D s + K_P \quad (30)$$

as

$$K_P = 120 \text{ voltage/degree}$$

$$K_D = 20 \text{ voltage/degree/s}$$

To implement this controller in SIMULINK, a filter was added to the derivate term. In other words, the derivative portion is not implemented as $K_D s$, but instead as

$$\frac{K_D s}{\tau s + 1}$$

where $\tau = 0.01$ s. This approach was taken so that the derivative portion of the controller is realized with a proper transfer function (total number of zeros is equal or less than the total number of poles in the transfer function). This also reflects standard

industry practices since signals are not perfect and need to be filtered before differentiation (this mitigates detrimental impact from signal noise). The resulting root locus plot when adding this filter is given in Figure 4.1 (notice the pole at -3.14).

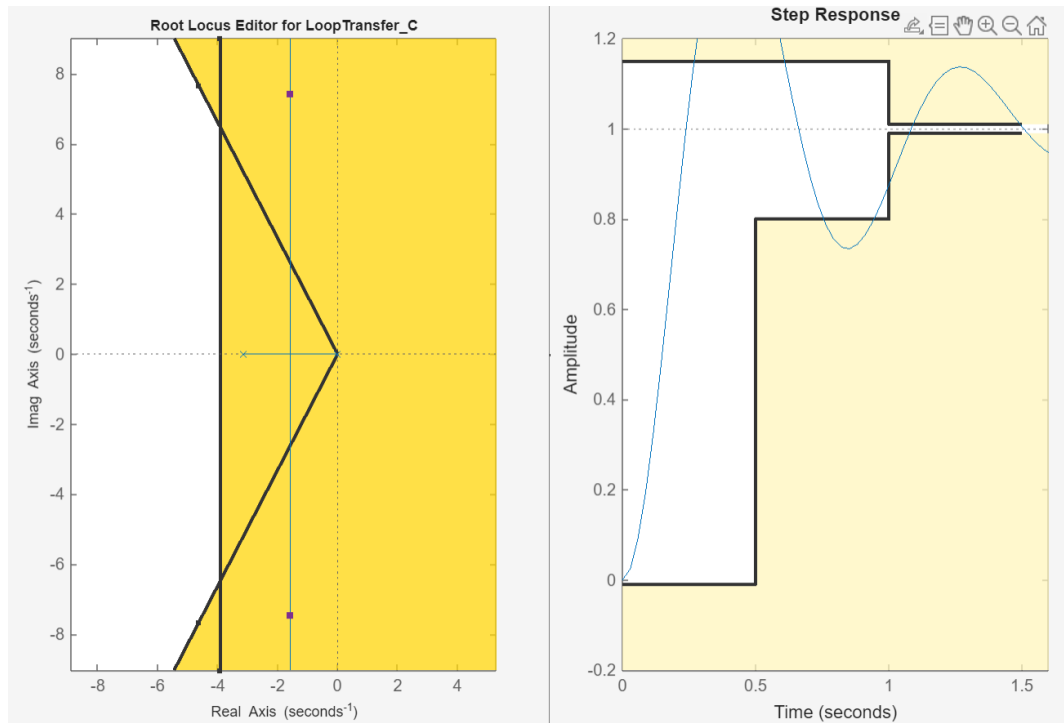


Figure 4.2 Root locus and response curve when using only proportional gain.

A simulation was attempted using only a proportional controller, however, this led to a system which did not align with the specified parameters. Therefore, a PD controller was the better option. The complete closed-loop control system implemented in SIMULINK is given in Figure 4.3. The plant parameters and controller parameters are imputed in the SIMULINK model and loaded into MATLAB's workspace prior to running a simulation. This approach is taken to simplify evaluations of plant parameter variations.

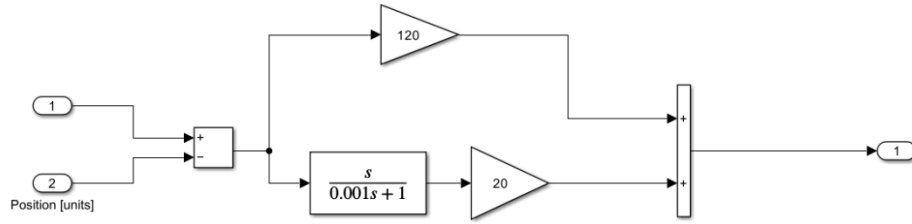


FIGURE 4.5 SIMULINK IMPLEMENTATION OF CLOSED-LOOP SYSTEM (PD ARCHITECTURE).

5 PERFORMANCE EVALUATION

The performance of the control system was evaluated against the specifications outlined in Table 2.3. This evaluation was conducted using a baseline set of plant parameters in the simulation, as well as under conditions where the drone weight and total inertia were varied. These were the only parameters altered from those listed in Table 2.1, allowing the assessment to focus solely on the impact of mass and inertia changes. This approach ensured that the system's ability to meet performance requirements was thoroughly tested under realistic variations in physical conditions.

5.1 AS DESIGNED

The results of the full closed-loop control system simulation, developed in SIMULINK and illustrated in Figure 4.3, are presented taking no mass added onto the system in Figure 4.1, and mass of the drone added to the drone in Figure 4.2. This simulation uses the nominal plant parameters listed in Table 2.1. Table 5.1 provides performance metrics from the simulation to confirm that the design meets the specified requirements in both scenarios, the arm working with and without a drone. The performance specifications are concentrated about θ_2 , where the arm is at 120 degrees.

TABLE 0.1 Compliance verification for performance specifications.

Parameter	Specification	Simulation	Units	P/F
Steady state error at θ_2	<0	0.0	degrees	pass
Overshoot at θ_2	<15	7.19%	%	pass
Settling time at θ_2	<1s	0.87	s	pass

5.2 PERTURBATION & ROBUSTNESS EVALUATION

To analyze the robustness of the control system for the backpack-mounted robotic arm, the key plant parameters—drone mass (M) and total inertia (J_t)—were varied within their expected tolerances. Three extreme cases were evaluated to represent the limits of physical variation: in Case 1, the drone mass was set to 0 kg, with inertia scaled accordingly to reflect a minimal-load scenario; in Case 2, the mass was increased to 1 kg, with total inertia proportionally adjusted; and in Case 3, the mass was further increased to 2 kg, again with inertia scaled to match. These cases were selected to assess how the system performs under both favorable and extreme dynamic conditions. Simulation results, overlaid in Figure 5.2, 5.4, and 5.6, confirm that variations in plant parameters do not significantly impact system performance. All control objectives and performance specifications remained satisfied across the tested scenarios, demonstrating that the system is robust to the modeled uncertainties.

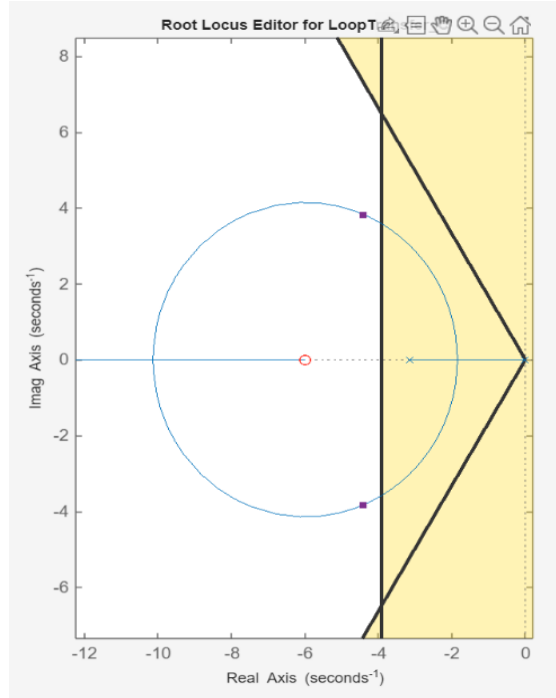


FIGURE 5.1 ROOT LOCUS FOR **CASE 1**, WHERE THE MASS OF THE DRONE IS NOT CONSIDERED.

Transfer function, shown in equation 31, is used to simulate the mass added by the drone by adding its inertia to J_2 . In this case, there is no drone mass, (M), is applied and so $J_2 = 0.08723$

$$G_p = \frac{\theta_2(s)}{V_I(s)} = \frac{6.2}{12.87s^2 + 40.35s} \quad (31)$$

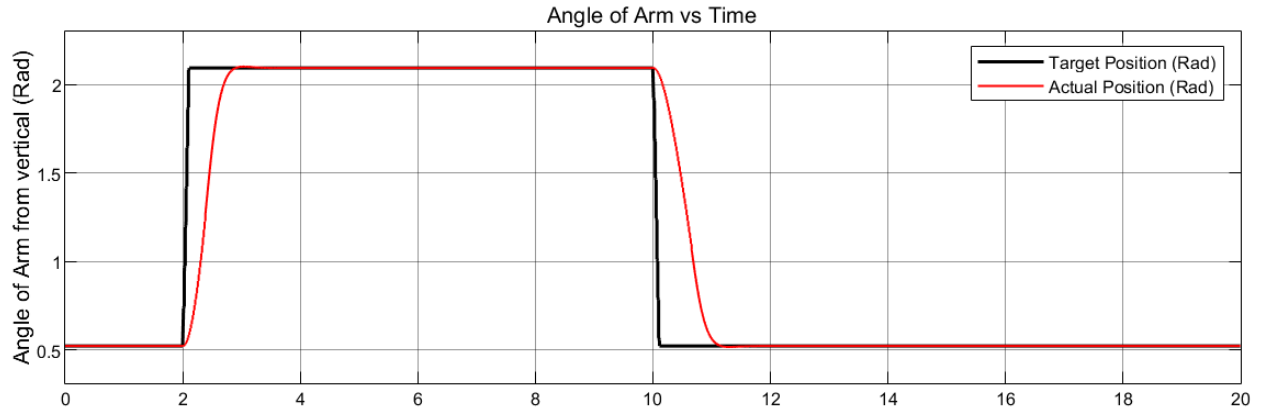


FIGURE 5.2 RESPONSE CURVE FOR **CASE 1**, ARM MOVES WITHOUT A DRONE ADDED.

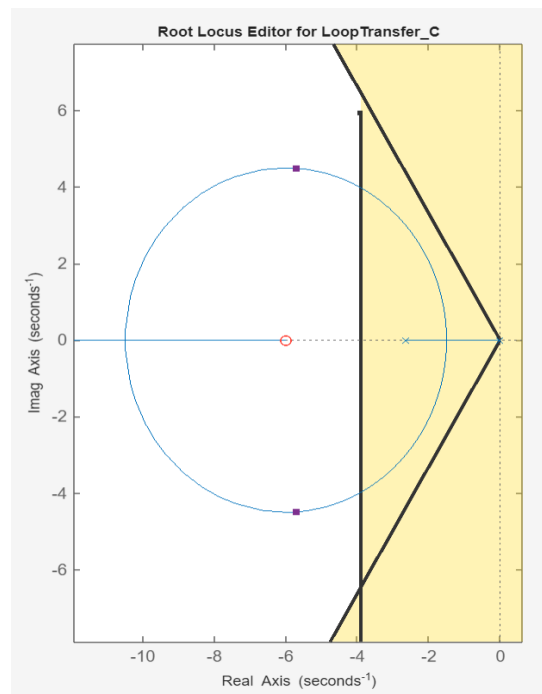


FIGURE 5.3 ROOT LOCUS OF **CASE 2**, WHERE THE MASS OF THE DRONE IS 1 KG

Transfer function, shown in equation 32, is used to simulate the mass added by the drone by adding its inertia to J_2 . In this case, the mass, (M), of the drone is 1kg is applied and so $J_2 = 0.33723$

$$G_p = \frac{\theta_2(s)}{V_I(s)} = \frac{6.2}{15.3723s^2 + 40.35s} \quad (32)$$

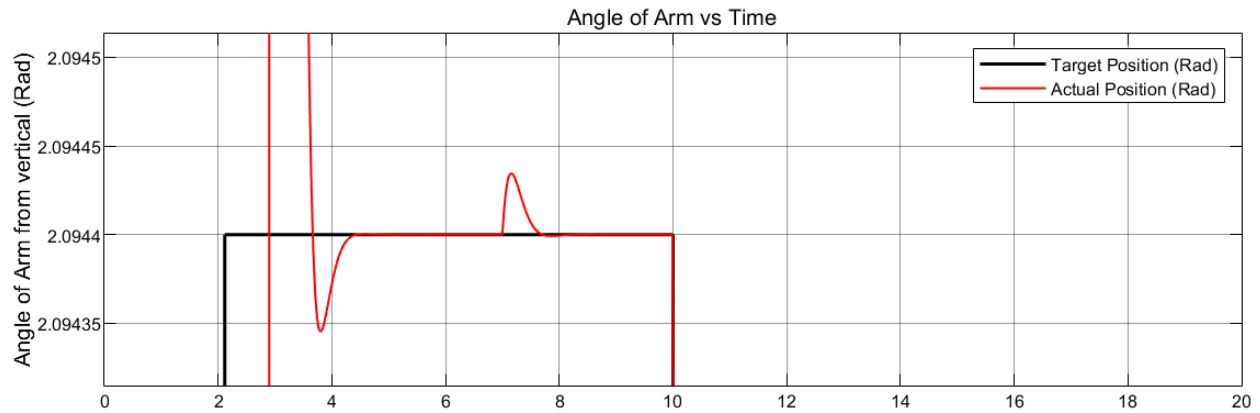


FIGURE 5.4 RESPONSE CURVE FOR CASE 2. SHOWS ADDED WEIGHT IMPACTING THE ANGLE OF ARM.

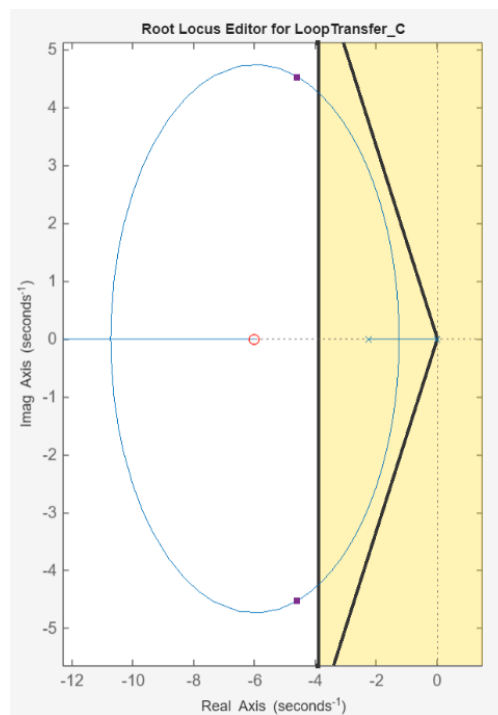


FIGURE 5.5 ROOT LOCUS FOR CASE 3, WHERE THE MASS OF THE DRONE IS 2 KG.

Transfer function in equation 33 is used in the above root locus to simulate the mass added by the drone by adding its inertia to J_2 .

$$G_p = \frac{\theta_2(s)}{V_I(s)} = \frac{6.2}{17.87s^2 + 40.35s} \quad (33)$$

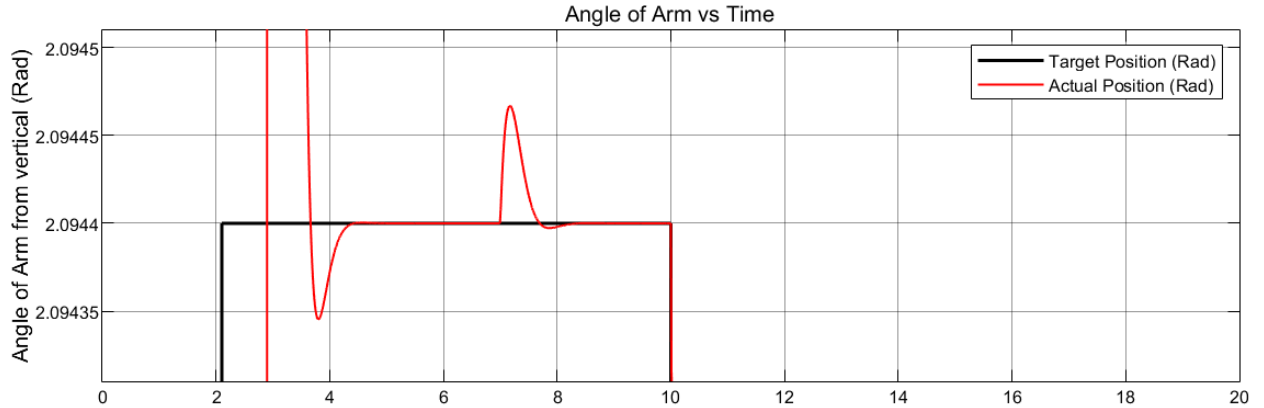


FIGURE 5.6 RESPONSE CURVE FOR **CASE 3**. SHOWS ADDED WEIGHT IMPACTING THE ANGLE OF ARM.

6 CONCLUSIONS

The goal of this project was to design and analyze a closed-loop control system for a backpack-mounted robotic arm intended to retrieve and deploy a drone. This was initially attempted using an open-loop response. However, this revealed that the open-loop response was inadequate due to major limitations, such as the inability to reliably position the arm. The open loop could not reach the desired position reliably and had a large steady state error. This showed that a closed-loop response was required for the desired functionality. Using MATLAB and SIMULINK, a nonlinear closed-loop model of the system was developed. A PD-based control system was then designed via root locus methods to meet the performance specifications. PD controller is used over just a P gain because no gains would place us in the required specification range. The PD controller was sufficient considering the transfer function contained a natural integrator. The performance specifications were set for the system to a settling time of no more than one second, a maximum overshoot of 15%, and a steady-state error of 0 degrees. The results of the simulations revealed that the new system was stable and robust under normal conditions. These simulations also proved that the system stays within specifications through some different cases where key variables were changed, such as the inertia changing due to different drone masses. Certain assumptions were made to simplify the modeling process, such as neglecting some non-linear disturbances and simplifying the effect of the drone. However, this can be fixed in the future through refining the mechanical model, additional simulation, or physical implementation and testing under real-world

conditions. This project establishes a strong foundation for a functional and reliable drone retrieval system that can be extended and optimized for future applications.

7 REFERENCES

- [1] N. Nise, Control Systems Engineering, 4th ed., Hoboken NJ: John Wiley & Sons Inc., 2004.

APPENDIX

ADDITIONAL DATA

A.1 Ramp function implemented in the control system is a D-1 lookup table so that the voltage response from the controller is more realistic and doesn't have unrealistic spikes and keeps the system acting more predictably.

A-2 The equation used for calculating inertias is $J=m \cdot r^2$, where J is the variable for inertia, m is the variable for mass, and r is the radius of rotation or the length.